

Sales Performance Dashboard

1. DATASET DETAILS

Source of Data: Kaggle – Global Superstore Dataset

Number of Records/Samples: 9,994 Rows

Features / Attributes (Columns):

Column Name Data Type

Order ID	String
Customer ID	String
Order Date	Date
Ship Date	Date
Product Name	String
Category	String
Sub-Category	String
Region	String
State	String
City	String
Segment	String
Ship Mode	String
Sales	Numeric
Profit	Numeric
Discount	Numeric
Quantity	Numeric

Training / Testing Split: Not Applicable (Business Intelligence & Descriptive Analytics Project)

Data Format: CSV

2. ALGORITHM

Step-by-Step Procedure

Step 1: Data Collection

Import the Global Superstore Dataset (CSV) into the Python environment.

Step 2: Data Cleaning & Preprocessing

Use Pandas and NumPy to remove duplicate records, handle missing values, standardize column names, and convert data into the required format.

Step 3: Exploratory Data Analysis (EDA)

Analyze sales trends, profit trends, customer purchasing behavior, seasonal patterns, and outliers using Matplotlib and Seaborn.

Step 4: KPI Calculation

Calculate:

- Total Revenue
- Total Profit
- Total Orders
- Average Order Value
- Sales Growth
- Top Performing Products
- Top Performing Regions

Step 5: Data Storage

Store the cleaned dataset in a SQL database for structured querying and efficient data retrieval.

Step 6: Dashboard Development

Connect Power BI with the SQL database and build interactive dashboards using KPI cards, charts, tables, slicers, and filters.

Step 7: Reporting

Generate weekly and monthly reports to monitor business performance and support decision-making.

Logic / Method Used

Business Intelligence (BI) methodology using the ETL (Extract, Transform, Load) process to convert raw sales data into meaningful business insights.

Model / Technique Applied

- Business Intelligence (BI)
- Exploratory Data Analysis (EDA)
- Statistical Aggregation
- Time-Series Trend Analysis

- Interactive Dashboard Visualization

Process Flow of the System

Raw Sales Data → Data Cleaning → Data Analysis → KPI Calculation → SQL Database → Power BI Dashboard → Business Insights

3. IMAGINATION OF DASHBOARDS

Dashboard 1 – Executive Overview

This dashboard provides a complete overview of organizational sales performance. It includes KPI cards displaying **Total Revenue, Total Profit, Total Orders, and Average Order Value**. A line chart shows monthly sales and profit trends, a bar chart displays the top-performing products, and a donut chart presents regional sales distribution. Interactive Date and Region filters allow users to analyze business performance efficiently.

Dashboard 2 – Regional & Product Analysis

This dashboard helps managers analyze sales performance across different regions and product categories. It includes a filled map showing state-wise sales, a stacked bar chart comparing Technology, Furniture, and Office Supplies categories, a matrix visual for region-wise sales comparison, and a detailed table displaying product-wise sales and profit with conditional formatting. Filters for State, City, and Category enable detailed analysis.

Dashboard 3 – Customer & Sales Insights

This dashboard focuses on customer behavior and sales performance. It contains a clustered bar chart showing revenue by Customer Segment (Consumer, Corporate, and Home Office), a scatter chart illustrating the relationship between Discount and Profit, a KPI card highlighting the highest revenue-generating order, and a table listing the Top 20 Customers based on total revenue. These insights help organizations optimize discount strategies, improve customer satisfaction, and increase profitability.

4. VISUALIZATION ELEMENTS USED

Charts

- Line Chart
- Bar Chart
- Stacked Column Chart
- Donut Chart
- Scatter Plot
- Filled Map
- Matrix Visual

KPI Elements

- KPI Cards

- Gauge Charts

Interactive Filters

- Date Slicer
- Region Filter
- Category Filter
- Customer Segment Filter

Tables

- Product-wise Sales Table
- Region-wise Sales Table
- Customer-wise Sales Table

Features

- Drill-through
- Tooltips
- Conditional Formatting
- Export to PDF
- Export to Excel

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